Science Olympiad
Astronomy C
Purdue Invitational

January 18, 2025

Directions:

- Each team will be given **50** minutes to complete this test.
- This test can be taken apart, but should not be written on **at all**. All questions should be answered on the provided answer sheet.
- There are three sections: **A** (Multiple Choice), **B** (DSOs), and **C** (Free Response). These sections can be completed in any order.
- Question C15 is the tiebreaker question. The aesthetics and overall quality of that sketch as well the inclusion of additional detail will be used to break ties.
- Appropriate units should be provided with all answers.
- Good luck!

Written by:
Rohan Kelkar, *kelkar5@purdue.edu*
Connor Siurek, *csiurek@purdue.edu*

Section A: Multiple Choice (25 points)

All questions are worth one point. Only the letters 'A' through 'D' should be written on the answer sheet. 'T' and 'F' will NOT be accepted.

1. Brown dwarfs are similar in density to white dwarfs.
   (a) True
   (b) False
2. Blueshift refers to the shift in spectral lines towards longer wavelengths in radiation from distant galaxies and celestial objects.
   (a) True
   (b) False
3. A kilonova is an extremely energetic supernova which results from the core collapse of a star of over 30 solar masses.
   (a) True
   (b) False
4. Brown dwarfs are unable to sustain fusion.
   (a) True
   (b) False
5. The Earth Similarity Index can show how habitable a planet is.
   (a) True
   (b) False
6. The fictional planet Krootabulon-A is located 1AU away from its star, recieving light with intensity of I W/m^2. What is the the intensity of light recieved of a planet Krootabulon-B, which is located only 0.5AU away from the same star?
   (a) I/4
   (b) I/2
   (c) 2I
   (d) 4I
7. Which part of a galaxy contains the majority of dark matter?
   (a) Disk
   (b) Bulge
   (c) Arms
   (d) Halo
8. If a star is observed to have weak hydrogen lines, which spectral class does it likely fall into?
   (a) O
   (b) A
   (c) K
   (d) M
9. What is the next stage in our Sun's evolution and in rougly how many years will this occur?
   (a) Red giant, 500 million years
   (b) Red giant, 5 billion years
   (c) White dwarf, 500 million years
   (d) White dwarf, 5 billion years
10. Which of the following is not one of the possible end stages of a star with mass of at least 8 solar masses?
    (a) Planetary nebula
    (b) Black hole
    (c) Neutron star
    (d) Pulsar
11. Which super-Earth was the first discovered with very similar conditions to Earth?
    (a) Gilese 581c
    (b) Kepler-10b
    (c) MOA-2007-BLG-192Lb
    (d) Kepler-452b
12. Wolf Rayet stars are NOT identified by strong emission lines of which of the following element?
    (a) Nitrogren
    (b) Beryllium
    (c) Silicon
    (d) Carbon
13. Which of the following is a supermassive star with a very short lifetime and unpredictable outbursts?
    (a) Luminous blue variables
    (b) Mira variables
    (c) Neutron stars
    (d) T Tauri stars

14. Which of the following is a reasonable value for the number of stars in a globular cluster?
    (a) 10^2
    (b) 10^4
    (c) 10^6
    (d) 10^8

15. Which of the following is NOT a characteristic associated with Herbig Ae/Be stars?
    (a) Hydrogen fusing
    (b) Calcium emission lines
    (c) Circumstellar disks
    (d) Brightness variability

16. Which of the following is a difference between Population I and Population II stars?
    (a) Population I stars were formed at an earlier period in the universe
    (b) Population II stars have only been theorized, but not actually observed.
    (c) Population II stars are far more massive than Population I stars
    (d) Population I stars have a higher metallicity than Population II

17. In the Sun-Moon system, what is the approximate distance from the 1st Lagrange point (L_1) and the 2nd Lagrange point (L_2)?
    (a) 0 AU
    (b) $1 \cdot 10^{-4}$ AU
    (c) $1 \cdot 10^{-2}$ AU
    (d) 1 AU

18. In the Sun-Earth system, what is the approximate distance between the Earth and the 3rd Lagrange point?
    (a) 0.5 AU
    (b) $\frac{\sqrt{3}}{2}$ AU
    (c) $\frac{\sqrt{2}}{2}$ AU
    (d) 2 AU

19. Which method of exoplanet detection can reveal the mass of the planet?
    (a) Transit
    (b) Radial velocity
    (c) Direct imaging
    (d) Gravitational microlensing

20. A sub-Earth is a planet with substantially less what?
    (a) Heat
    (b) Volume
    (c) Mass
    (d) Nitrogen in its atmosphere

21. In a globular cluster with a half-mass radius of 33 ly, stars are interacting via gravitational encounters. What happens to the central density and velocity dispersion of the stars if the cluster undergoes core collapse?
    (a) The central density decreases and velocity dispersion increases
    (b) The central density increases and velocity dispersion decreases
    (c) The central density increases and velocity dispersion increases
    (d) The central density decreases and velocity dispersion decreases

22. What kind of stars have the most uniform chemical composition?
    (a) Zero-age main sequence (ZAMS)
    (b) White dwarfs
    (c) Hypergiants
    (d) Neutron stars

23. From what other formula or Law can you derive Wien's law?
    (a) Schrödinger's equation
    (b) Stefan-Boltzmann law
    (c) Hubble's law
    (d) Planck's law

24. What are T Tauri stars powered by?
    (a) Gravitational contraction
    (b) Hydrogen fusion
    (c) Deuterium fusion
    (d) Carbon fusion

25. What is a reasonable value for the lifetime of a red supergiant?
    (a) 5 million years
    (b) 50 million years
    (c) 500 million years
    (d) 5 billion years

Section B: DSOs (30 points)

The following questions will refer to this year's DSO list and the image sheet. Point values are denoted on each question.

1. [1 pt] What is the mass of LHS 3844b in kg?
2. [1 pt] Image B is an artist's vizualization of which DSO?
3. [1 pt] What is the satellite that dicovered the exoplanet LTT 9779 b?
4. [1 pt] What is the approximate age of Kepler 62?
5. [1 pt] What constellation is exoplanet depicted in image E in?
6. [2 pt] Image G is composed from a number of images taken in what wavelength?
7. [2 pts] WASP 17-b is the first planet to be found to orbit in a direction counter to the rotation of it's host star. What is this phenomenon called?
8. [2 pts] There are two unique facts about the planetary system of PSR B1257+12. Name them both.
9. [5 pts] Refer to image A
   (a) [1 pt] Which DSO is pictured?
   (b) [1 pt] What type of object is pictured?
   (c) [1 pt] What is the name of the other object in the image?
   (d) [1 pt] What does the close distance between the two objects suggest about the formation of the system?
   (e) [1 pt] What is a proposed theory for the answer to part (d)?
10. [5 pts]
    (a) [1 pt] Which image depicts 30 Doradus?
    (b) [1 pt] What is the most common nickname for this DSO?
    (c) [1 pt] What are the superimposed lines on this image?
    (d) [2 pts] What are two functions of the answer in part (c) for this DSO?
11. [5 pts]
    (a) [1 pt] Which image depicts the Orion Nebula?
    (b) [1 pt] Provide one other name for the nebula.
    (c) [1 pt] Which instrument took this image?
    (d) [2 pts] What is the name for the object in the center of this image?
12. [4 pts] Refer to image C
    (a) [1 pt] What is the DSO that is pictured at the center?
    (b) [1 pt] What is the ring highlighted around the central object?
    (c) [2 pts] What is the name of the effect that creates this ring?

Section C: Free Response (45 points)

Point values are denoted on the individual questions.

Short Answer

(Questions should be answered in a few words or numbers.)

1. [1 pt] Gravitational waves travel at what speed?
2. [1 pt] What is the Chandrasekhar limit?
3. [1 pt] What is the controlling factor of a star's evolutionary track?
4. [1 pt] Where do protostars form?
5. [1 pt] Which method of exoplanet detection measures Doppler shifts in the spectrum of the parent star?
6. [2 pts] What is the most successful method of exoplanet detection? Rougly what percent of all exoplanets were detected this way?
7. [2 pts] What are two ways that newly formed planets obtain heat?
8. [1 pt] Besides intenal heating, what is another theorized way that ice planets have subsurface oceans?
9. [1 pt] According to the original conception of the "habitable zone" in 1959, why should large-mass bodies not be present in or near a system's habiltability zone?

Open Response

(Questions will likely require multiple sentences and/or work shown to succcesfully earn all points.)

10. [4 pts] Which Lagrange point does the James Webb Space Telescope orbit in? Provide 3 reasons why NASA decided to use that Lagrange point.
11. [3 pts] A buddy of mine who weighs 60 kg was bored one night and bought a paperweight off of Temu. The paperweight is a cube with a side length 1 cm and the density of a neutron star. What is the force, in newtons, applied by this cube on him when he is 1 meter away?
12. [4 pts] If the length of a habitability zone is 9.62 AU what is the inner, outer, and middle radius?

Use image F to respond to question 13

13. [8 pts] Let's say we observe a main sequence star with a mass of 4 $M_\odot$ and it emits a wavelength of around 264 nm. The exoplanet's radius is 3 % of its star's radius, the planet is on the center of the stellar disk, and it orbits roughly 5 AU away from the star. Use the fact that the planet's mass does not affect calculations to answer the following questions.
    (a) [2 pts] What is the transit duration in days?
    (b) [2 pts] What is the orbital velocity of the planet in km/s?
    (c) [2 pts] If the albedo of this exoplanet is 0.064, what is the equilibrium temperature of the exoplanet, in Kelvin?
    (d) [1 pt] Will water boil, assuming that atmospheric pressure is 1 atm?
    (e) [1 pt] If the planet moves counter-clockwise in the plane of the page, what direction does the total angular momentum point?

14. [7 pts] Imagine that you can look at the process that happens within a distant star. A Balmer series occurs where electrons seem to come from the largest possible excited state. Assume the production of hydrogen is mainly occurring, and use the wavelength that the Balmer series is converging towards for calculations.

    (a) [2 pts] What is the temperature of the star in Kelvin?

    (b) [1 pt] What spectral classification is the star?

    While looking at the star, you've realized that the observed wavelength is 372.95nm.

    (c) [2 pts] At what velocity is the star moving away from the observer in km/s?

    (d) [2 pts] How far away is the star from the observer in Mpc?

15. [8 pts, tiebreaker!!!] Sketch an Hertzsprung-Russell diagram. Draw the following regions: white dwarfs, main sequence, giants, and supergiants. Label the Sun on your diagram with its spectral classification. Make sure that your diagram is titled and that axes are present and labeled.